Impedance Control of a 2-DOF Robot Arm

Ashvin Anilkumar

***Abstract*— We study joint-space impedance control of a two-link arm moving in a vertical plane. We derive the nonlinear model and its state-space form, linearize it about the hanging equilibrium, and test controllability, observability, stabilizability and detectability at every equilibrium and with actuators or sensors removed. The impedance law, gravity compensation plus a virtual spring and damper, turns the closed loop into a mass-spring-damper whose stiffness and damping we choose. An energy function proves global asymptotic stability and shows that the closed loop is passive with respect to external torques. Because the arm's inertia differs by a factor of up to 34 between directions, a single damping gain cannot damp both modes. A damping matrix built from the square root of the inertia matrix gives every mode the same damping ratio. In nonlinear simulation this design settles a 1.22 rad step in 2.6 s with 1.3% overshoot, against 7.0 s and 13% for the original gains, at the cost of a larger peak deflection under a sharp push.**

I. Introduction

A robot that works next to people, or that touches its surroundings, cannot simply follow positions. A stiff position controller pressed against a rigid surface turns a small position error into a large contact force. *Impedance control* [1] regulates the relation between motion and force instead: the robot responds to a push like a mass-spring-damper chosen by the designer. The simplest form needs only joint measurements. It is a proportional-derivative law with gravity compensation, and its global stability for rigid manipulators goes back to Takegaki and Arimoto [2].

We study this controller on the 2-DOF arm in Fig. 1. Its equations of motion are nonlinear (sines, cosines and products of velocities), yet small enough that every matrix can be written out and checked. Throughout, we carry one concrete example arm, and every number in the paper is reproduced by a script that simulates the full nonlinear model. The contributions are:

- a state-space model, its linearization, and the open-loop behavior (Sec. III);
- a structural analysis: controllability, observability, stabilizability and detectability at all four equilibria, with one actuator or one sensor removed, and without gravity, together with a condition for when a single joint motor misses a mode (Sec. III-D);
- the impedance law with its static stiffness, a global stability proof, passivity, and the local eigenvalues (Sec. IV);
- an explanation of why scalar damping gains cannot damp both modes, and a damping matrix that gives every mode the same damping ratio (Sec. V);
- nonlinear simulations that test each prediction, including the effect of an error in the gravity model (Sec. VI).

II. System, Inputs & Notation

A. Example arm

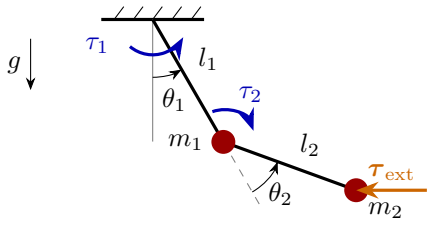


Fig. 1. 2-DOF arm (adapted from [3]). Joint angles θ_1, θ_2 are zero when the arm hangs straight down; θ_2 is relative to link 1. The controller commands the motor torques τ_1 (shoulder) and τ_2 (elbow) in blue. The environment acts through the external torque $\boldsymbol{\tau}_{\text{ext}}$ (orange), which we cannot command.

For all numerical examples we use point masses at the link tips with $m_1 = m_2 = 1$ kg, $l_1 = l_2 = 1$ m, and $g = 9.81$ m/s^2.

B. How we interact with the arm

The arm has two actuators, one at each joint. Our *control input* is the vector of motor torques

$$\mathbf{u} = [\tau_1 \quad \tau_2]^T \in \mathbb{R}^2,$$

where τ_1 acts at the shoulder (on link 1) and τ_2 at the elbow (between link 1 and link 2). Because each joint has its own motor, both degrees of freedom are directly actuated. The environment (contact, or a push on the arm) acts through the *external torque* $\boldsymbol{\tau}_{\text{ext}} \in \mathbb{R}^2$. It is a disturbance: we do not choose it, and it is zero in free motion. We assume we measure the joint angles and velocities, so the controller sees the full state. In the impedance controller of Section IV, the only other signal we supply is the desired joint configuration $\boldsymbol{\theta}_d$, which sets the equilibrium the virtual spring pulls toward.

C. Notation

Table I lists every symbol used in this paper. Vectors and matrices are bold; a dot is a time derivative.

D. Control objective

Given a constant desired configuration $\boldsymbol{\theta}_d$, design a state-feedback law $\mathbf{u} = \boldsymbol{\pi}(\mathbf{x}, \boldsymbol{\theta}_d)$ such that (i) with $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$ the error $\mathbf{e} = \boldsymbol{\theta} - \boldsymbol{\theta}_d$ converges to zero, and (ii) when $\boldsymbol{\tau}_{\text{ext}} \neq \mathbf{0}$ the arm yields like a spring-damper, $\mathbf{K}\mathbf{e} = \boldsymbol{\tau}_{\text{ext}}$ at rest, with $\mathbf{K}$ and $\mathbf{D}$ chosen by us.

TABLE I

SYMBOLS AND EXAMPLE VALUES.

Symbol	Meaning	Example
θ_1, θ_2	joint angles (shoulder, elbow); $\theta = 0$ is hanging down	rad
$\dot{\boldsymbol{\theta}}, \ddot{\boldsymbol{\theta}}$	joint velocity, acceleration	rad/s, rad/s^2
$m_i,\ l_i$	mass and length of link i	1 kg, 1 m
g	gravitational acceleration	9.81 m/s^2
$\mathbf{M}(\boldsymbol{\theta})$	inertia matrix	2×2
$\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$	Coriolis/centrifugal matrix	2×2
$\boldsymbol{\tau}_g(\boldsymbol{\theta})$	gravity torque vector	N·m
$\mathbf{u} = [\tau_1\ \tau_2]^T$	motor torques (control input)	N·m
$\mathbf{B}$	maps $\mathbf{u}$ to joint torques; $\mathbf{I}_2$ here	2×2
$\boldsymbol{\tau}_{\text{ext}}$	external (contact) torque, uncommanded	N·m
$\mathbf{x}$	state $[\boldsymbol{\theta}^T\ \dot{\boldsymbol{\theta}}^T]^T$	$\mathbb{R}^4$
$\mathbf{f}(\mathbf{x}, \mathbf{u})$	state-space dynamics, $\dot{\mathbf{x}} = \mathbf{f}$	
$(\mathbf{x}^*, \mathbf{u}^*)$	equilibrium state–input pair	$(\mathbf{0}, \mathbf{0})$
$\mathbf{A}, \mathbf{B}_{\text{lin}}$	Jacobians of $\mathbf{f}$ at $(\mathbf{x}^*, \mathbf{u}^*)$	4×4, 4×2
$\boldsymbol{\theta}_d,\ \mathbf{e}$	desired angles; error $\boldsymbol{\theta} - \boldsymbol{\theta}_d$	rad
$\mathbf{K}, \mathbf{D}$	virtual stiffness and damping gains	2×2

III. MATHEMATICAL MODEL & LINEARIZATION

A. Dynamics

As given by [3], the equations of motion (EOM) for a general manipulator with joint angles $\boldsymbol{\theta} \in \mathbb{R}^n$ are

$$\boxed{\mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} = \boldsymbol{\tau}_g(\boldsymbol{\theta}) + \mathbf{B}\mathbf{u} + \boldsymbol{\tau}_{\text{ext}}.} \tag{1}$$

Compared with [3], the term $\boldsymbol{\tau}_{\text{ext}}$ is added so that interaction with the environment can be studied. For our arm ($n = 2$), with $c_2 = \cos\theta_2$, $s_1 = \sin\theta_1$, $s_2 = \sin\theta_2$, $s_{12} = \sin(\theta_1 + \theta_2)$,

$$\mathbf{M}(\boldsymbol{\theta}) = \begin{bmatrix} (m_1 + m_2)l_1^2 + m_2 l_2^2 + 2m_2 l_1 l_2 c_2 & m_2 l_2^2 + m_2 l_1 l_2 c_2 \\ m_2 l_2^2 + m_2 l_1 l_2 c_2 & m_2 l_2^2 \end{bmatrix}$$

$$\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) = \begin{bmatrix} 0 & -m_2 l_1 l_2 (2\dot{\theta}_1 + \dot{\theta}_2) s_2 \\ \frac{1}{2} m_2 l_1 l_2 (2\dot{\theta}_1 + \dot{\theta}_2) s_2 & -\frac{1}{2} m_2 l_1 l_2 \dot{\theta}_1 s_2 \end{bmatrix}$$

$$\boldsymbol{\tau}_g(\boldsymbol{\theta}) = -g \begin{bmatrix} (m_1 + m_2) l_1 s_1 + m_2 l_2 s_{12} \\ m_2 l_2 s_{12} \end{bmatrix}, \qquad \mathbf{B} = \mathbf{I}_2,$$

so both joints are directly actuated. With the example values ($m_i = l_i = 1$) these become

$$\mathbf{M} = \begin{bmatrix} 3 + 2c_2 & 1 + c_2 \\ 1 + c_2 & 1 \end{bmatrix}, \quad \boldsymbol{\tau}_g = -g \begin{bmatrix} 2s_1 + s_{12} \\ s_{12} \end{bmatrix}.$$

Remark 1 *The matrix $\mathbf{C}$ above reproduces the correct Coriolis force $\mathbf{C}\dot{\boldsymbol{\theta}}$, but it is not the Christoffel-symbol factorization, so $\dot{\mathbf{M}} - 2\mathbf{C}$ is not skew-symmetric. This does not affect the equations of motion, the linearization, or the simulations below. The stability proof of Sec. IV needs only the scalar identity $\dot{\boldsymbol{\theta}}^T(\dot{\mathbf{M}} - 2\mathbf{C})\dot{\boldsymbol{\theta}} = 0$, which every valid $\mathbf{C}$ satisfies. An argument that applies $\dot{\mathbf{M}} - 2\mathbf{C}$ to other vectors must instead use $\mathbf{C}_{\text{ch}} = \begin{bmatrix} -h\dot{\theta}_2 & -h(\dot{\theta}_1 + \dot{\theta}_2) \\ h\dot{\theta}_1 & 0 \end{bmatrix}$ with $h = m_2 l_1 l_2 s_2$, which gives the same $\mathbf{C}_{\text{ch}}\dot{\boldsymbol{\theta}}$ and satisfies $\dot{\mathbf{M}} - 2\mathbf{C}_{\text{ch}}$ skew-symmetric. We verified both facts numerically.*

Second-order form. For strictly positive masses and link lengths, $\mathbf{M}(\boldsymbol{\theta})$ is positive definite and therefore invertible, so the EOM solve explicitly for the joint accelerations:

$$\ddot{\boldsymbol{\theta}} = \mathbf{M}(\boldsymbol{\theta})^{-1} \left[-\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} + \boldsymbol{\tau}_g(\boldsymbol{\theta}) + \mathbf{B}\mathbf{u} + \boldsymbol{\tau}_{\text{ext}} \right]. \tag{2}$$

State-space form. Stacking $\mathbf{x} = [\theta_1\ \ \theta_2\ \ \dot{\theta}_1\ \ \dot{\theta}_2]^T = [\boldsymbol{\theta}^T\ \ \dot{\boldsymbol{\theta}}^T]^T$ and setting $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$, the dynamics $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ are, explicitly,

$$\mathbf{f}(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} \dot{\boldsymbol{\theta}} \\ \mathbf{M}(\boldsymbol{\theta})^{-1} \left[-\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} + \boldsymbol{\tau}_g(\boldsymbol{\theta}) + \mathbf{B}\mathbf{u} \right] \end{bmatrix}. \tag{3}$$

The measured output is the pair of joint angles, $\mathbf{y} = \mathbf{C}_y \mathbf{x}$ with $\mathbf{C}_y = [\mathbf{I}_2 \ \ \mathbf{0}_{2\times 2}]$.

B. Open-Loop Linearization

The arm hanging straight down at rest with zero torque is an equilibrium *state–input pair* $(\mathbf{x}^*, \mathbf{u}^*) = (\mathbf{0}, \mathbf{0})$: the velocities are zero and $\boldsymbol{\tau}_g(\mathbf{0}) = \mathbf{0}$, so $\mathbf{f}(\mathbf{x}^*, \mathbf{u}^*) = \mathbf{0}$. With $\delta\mathbf{x} = \mathbf{x} - \mathbf{x}^*$ and $\delta\mathbf{u} = \mathbf{u} - \mathbf{u}^*$, the first-order Taylor approximation is

$$\boxed{\delta\dot{\mathbf{x}} = \mathbf{A}\delta\mathbf{x} + \mathbf{B}_{\text{lin}}\delta\mathbf{u},} \tag{4}$$

$$\mathbf{A} = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{(\mathbf{x}^*, \mathbf{u}^*)}, \qquad \mathbf{B}_{\text{lin}} = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{u}}\right|_{(\mathbf{x}^*, \mathbf{u}^*)}.$$

Write the lower block of (3) as $\mathbf{M}^{-1}\mathbf{h}$ with $\mathbf{h} = -\mathbf{C}\dot{\boldsymbol{\theta}} + \boldsymbol{\tau}_g + \mathbf{B}\mathbf{u}$. Then $\partial(\mathbf{M}^{-1}\mathbf{h})/\partial\boldsymbol{\theta} = (\partial\mathbf{M}^{-1}/\partial\boldsymbol{\theta})\,\mathbf{h} + \mathbf{M}^{-1}\,\partial\mathbf{h}/\partial\boldsymbol{\theta}$, and the first term vanishes because $\mathbf{h}(\mathbf{x}^*, \mathbf{u}^*) = \mathbf{0}$. The Coriolis term $\mathbf{C}\dot{\boldsymbol{\theta}}$ is a sum of products $\dot{\theta}_i\dot{\theta}_j$ with coefficients depending on θ_2. Its derivative with respect to $\boldsymbol{\theta}$ is therefore proportional to $\dot{\theta}_i\dot{\theta}_j$, and its derivative with respect to $\dot{\boldsymbol{\theta}}$ is linear in $\dot{\boldsymbol{\theta}}$; both are zero at $\dot{\boldsymbol{\theta}} = \mathbf{0}$. Only gravity contributes, giving

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I}_2 \\ \mathbf{K}_g & \mathbf{0} \end{bmatrix}, \quad \mathbf{B}_{\text{lin}} = \begin{bmatrix} \mathbf{0} \\ \mathbf{M}_0^{-1}\mathbf{B} \end{bmatrix}, \quad \mathbf{K}_g = \mathbf{M}_0^{-1}\mathbf{G}, \tag{5}$$

where $\mathbf{M}_0 = \mathbf{M}(\mathbf{0})$ and $\mathbf{G} = \partial\boldsymbol{\tau}_g/\partial\boldsymbol{\theta}|_{\mathbf{0}}$, namely

$$\mathbf{M}_0 = \begin{bmatrix} (m_1 + m_2)l_1^2 + m_2 l_2^2 + 2m_2 l_1 l_2 & m_2 l_2 (l_1 + l_2) \\ m_2 l_2 (l_1 + l_2) & m_2 l_2^2 \end{bmatrix}, \ \mathbf{G} = -g \begin{bmatrix} (m_1 + m_2) l_1 + m_2 l_2 & m_2 l_2 \\ m_2 l_2 & m_2 l_2 \end{bmatrix}.$$

Example values. For $m_i = l_i = 1$ and $g = 9.81$: $\mathbf{M}_0 = \left[\begin{smallmatrix} 5 & 2 \\ 2 & 1 \end{smallmatrix}\right]$, $\mathbf{M}_0^{-1} = \left[\begin{smallmatrix} 1 & -2 \\ -2 & 5 \end{smallmatrix}\right]$, $\mathbf{G} = -g\left[\begin{smallmatrix} 3 & 1 \\ 1 & 1 \end{smallmatrix}\right]$, so

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -9.81 & 9.81 & 0 & 0 \\ 9.81 & -29.43 & 0 & 0 \end{bmatrix}, \quad \mathbf{B}_{\text{lin}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & -2 \\ -2 & 5 \end{bmatrix}.$$

The matrix $\mathbf{G}$ is symmetric negative definite and $\mathbf{M}_0$ is positive definite, so $\mathbf{K}_g = \mathbf{M}_0^{-1}\mathbf{G}$ is similar to the symmetric negative definite matrix $\mathbf{M}_0^{-1/2}\mathbf{G}\mathbf{M}_0^{-1/2}$ and has real negative eigenvalues $-\mu_i$. Each eigenvalue λ of $\mathbf{A}$ then satisfies $\lambda^2 = -\mu_i$, i.e. $\lambda = \pm j\sqrt{\mu_i}$. For the example, $\mu = g(2 \pm \sqrt{2})$ and the eigenvalues are $\pm j2.40$ and $\pm j5.79$ rad/s, which we confirmed numerically.

C. Open-Loop Analysis

Stability. All four eigenvalues of $\mathbf{A}$ lie on the imaginary axis, so the linearization is marginally stable: it oscillates at 2.40 and 5.79 rad/s and never settles, because the model has no dissipation. Such eigenvalues do not by themselves decide stability of the nonlinear system, but here energy conservation does: with $\mathbf{u} = \mathbf{0}$ the total energy is constant (drift below 10^{-8} J over 10 s in our simulation), so the arm swings indefinitely. This motivates a feedback law that adds damping.

Controllability. The controllability matrix $[\mathbf{B}_{\text{lin}} \;\; \mathbf{A}\mathbf{B}_{\text{lin}} \;\; \mathbf{A}^2\mathbf{B}_{\text{lin}} \;\; \mathbf{A}^3\mathbf{B}_{\text{lin}}] \in \mathbb{R}^{4\times 8}$ has rank 4. This also follows from structure: since $\mathbf{A}\mathbf{B}_{\text{lin}} = \begin{bmatrix} \mathbf{M}_0^{-1} \\ \mathbf{0} \end{bmatrix}$ and $\mathbf{M}_0$ is invertible, the first two blocks $[\mathbf{B}_{\text{lin}} \;\; \mathbf{A}\mathbf{B}_{\text{lin}}]$ alone have rank 4. With one motor per joint, every direction in $(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$ near the hanging equilibrium can be reached.

Observability. With $\mathbf{C}_y = [\mathbf{I}_2 \;\; \mathbf{0}]$, we have $\mathbf{C}_y\mathbf{A} = [\mathbf{0} \;\; \mathbf{I}_2]$, so the first two block rows of the observability matrix already form $\mathbf{I}_4$ and the rank is 4. The velocities can be reconstructed from measured angles, so an observer could replace velocity sensors. We assume full-state measurement and do not need one here.

D. Stabilizability and Detectability

By the PBH test [4], $(\mathbf{A}, \mathbf{B})$ is *stabilizable* iff $\operatorname{rank}[\lambda\mathbf{I} - \mathbf{A} \;\; \mathbf{B}] = 4$ for every eigenvalue with $\operatorname{Re}\lambda \geq 0$, and $(\mathbf{C}, \mathbf{A})$ is *detectable* iff the dual condition holds. Every open-loop eigenvalue of the hanging arm lies on the imaginary axis, so there are no asymptotically stable modes, and for this arm stabilizable is equivalent to controllable and detectable is equivalent to observable: losing any mode is fatal. Table II summarizes the test at all four equilibria $\boldsymbol{\theta}^* \in \{0, \pi\}^2$ and with actuators or sensors removed, computed from numerical Jacobians of the nonlinear model.

TABLE II

STRUCTURAL PROPERTIES OF THE LINEARIZATION (RANK OUT OF 4).

Case	Inputs / outputs	Ctrb./Obsv.	Stab./Det.
All 4 equilibria, $g = 9.81$	$\mathbf{u} = [\tau_1, \tau_2]$, $\mathbf{y} = \boldsymbol{\theta}$	4 / 4	yes / yes
All 4 equilibria, $g = 9.81$	τ_1 only or τ_2 only	4	yes
Hanging, $g = 9.81$	any single θ_i or $\dot{\theta}_i$	4	yes
Hanging, $g = 0$ (horizontal)	τ_1 only or τ_2 only	2	**no**
Hanging, $g = 0$ (horizontal)	$\mathbf{y} = \dot{\boldsymbol{\theta}}$, θ_1, or θ_2 only	2	**no**

Unstable equilibria. The upright configurations have eigenvalues in the open right half-plane (e.g. $\pm 2.40, \pm 5.79$ for $\boldsymbol{\theta}^* = (\pi, 0)$), but they are equally controllable and observable, so each can be stabilized locally by state feedback. Controllability is a property of $(\mathbf{A}, \mathbf{B})$, not of open-loop stability.

Role of gravity. With a single motor at joint j, $\mathbf{b}_j = [\mathbf{0}; \mathbf{M}_0^{-1}\mathbf{e}_j]$. If $g = 0$, $\mathbf{A}^2 = \mathbf{0}$ and the controllability matrix is $[\mathbf{b}_j \;\; \mathbf{A}\mathbf{b}_j]$ with rank 2. Two modes at $s = 0$ are lost, so the arm is not stabilizable. Likewise, velocity sensing alone cannot recover the absolute angles. With $g \neq 0$, $\mathbf{K}_g$ is invertible and couples the joints, so one motor, or one sensor of any kind, recovers rank 4.

When can a single motor lose a mode? Let the modes be $\mathbf{G}\mathbf{v} = -\mu\mathbf{M}_0\mathbf{v}$. The corresponding left eigenvector of $\mathbf{A}$ satisfies $\mathbf{w}^T\mathbf{b}_j \propto v_j$, and with $\mathbf{y} = \theta_j$ the right eigenvector gives $\mathbf{C}\mathbf{x} \propto v_j$. A motor or angle sensor at joint j therefore misses a mode iff that mode shape has no motion at joint j. Imposing $\mathbf{G}\mathbf{e}_1 \parallel \mathbf{M}_0\mathbf{e}_1$ gives $m_1 l_1 l_2 = 0$ (elbow only), and $\mathbf{G}\mathbf{e}_2 \parallel \mathbf{M}_0\mathbf{e}_2$ gives $l_1 = 0$ (shoulder only). For a real arm neither condition holds. As $m_1 \to 0$, however, the elbow-only margin $\min|v_2|/\|\mathbf{v}\|$ shrinks like $m_1/2$. The rigid "swing" mode is then barely excited by an elbow torque. At $m_1 = 0.01$ a numerical Kalman rank test already reports rank 2 although the system is controllable, so near-degenerate cases should be decided by the PBH or modal test.

Constructive stabilization. Linearizing the impedance law (6) at $\boldsymbol{\theta}_d = \mathbf{0}$ gives the state feedback $\delta\mathbf{u} = \mathbf{F}\delta\mathbf{x}$ with $\mathbf{F} = -[\mathbf{G} + \mathbf{K} \;\; \mathbf{D}]$, and $\mathbf{A} + \mathbf{B}_{\text{lin}}\mathbf{F} = \mathbf{A}_{cl}$ is Hurwitz (Sec. IV). This exhibits stabilizability directly. In closed loop the disturbance channel $(\mathbf{A}_{cl}, [\mathbf{0}; \mathbf{M}_d^{-1}])$ stays controllable, and for $\mathbf{K} = k\mathbf{I}$, $\mathbf{D} = d\mathbf{I}$ the position block of its controllability Gramian is $\mathbf{I}/(2dk)$, independent of $\mathbf{M}_d$.

IV. IMPEDANCE CONTROL

A. Control law and closed loop

Let $\boldsymbol{\theta}_d$ be a desired (constant) joint configuration. We apply the torque law

$$\boxed{\mathbf{u} = -\boldsymbol{\tau}_g(\boldsymbol{\theta}) + \mathbf{K}(\boldsymbol{\theta}_d - \boldsymbol{\theta}) - \mathbf{D}\dot{\boldsymbol{\theta}},} \tag{6}$$

where $\mathbf{K}, \mathbf{D} \in \mathbb{R}^{2\times 2}$ are symmetric positive definite stiffness and damping gains. The first term cancels gravity, and the other two act as a virtual spring and damper that pull the arm toward $\boldsymbol{\theta}_d$. Substituting into (1) gives the closed loop

$$\mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \Big(\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{D}\Big)\,\dot{\boldsymbol{\theta}} + \mathbf{K}\mathbf{e} = \boldsymbol{\tau}_{\text{ext}}, \qquad \mathbf{e} = \boldsymbol{\theta} - \boldsymbol{\theta}_d. \tag{7}$$

Static stiffness. If $\boldsymbol{\tau}_{\text{ext}}$ is constant and the arm comes to rest, the first two terms of (7) vanish and

$$\mathbf{K}\mathbf{e} = \boldsymbol{\tau}_{\text{ext}} \quad \Longleftrightarrow \quad \mathbf{e} = \mathbf{K}^{-1}\boldsymbol{\tau}_{\text{ext}}. \tag{8}$$

The deflection depends only on the chosen stiffness, not on the masses, the pose or gravity. This is part (ii) of the control objective in Sec. II-D.

B. Global stability and passivity

Proposition 1 *Let* $\mathbf{K}$ *and* $\mathbf{D}$ *be symmetric positive definite and* $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$*. Then the equilibrium* $(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) = (\boldsymbol{\theta}_d, \mathbf{0})$ *of* (7) *is globally asymptotically stable. For any* $\boldsymbol{\tau}_{\text{ext}}(t)$*, the function* V *below satisfies* $\dot{V} \leq \dot{\boldsymbol{\theta}}^T\boldsymbol{\tau}_{\text{ext}}$*.*

Proof. Take the kinetic energy plus the energy stored in the virtual spring,

$$V = \tfrac{1}{2}\dot{\boldsymbol{\theta}}^T\mathbf{M}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \tfrac{1}{2}\mathbf{e}^T\mathbf{K}\mathbf{e}. \tag{9}$$

Since $\det\mathbf{M} = m_2 l_1^2 l_2^2 (m_1 + m_2 \sin^2\theta_2) \geq m_1 m_2 l_1^2 l_2^2 > 0$ and the entries of $\mathbf{M}$ are bounded, the smallest eigenvalue of

$\mathbf{M}(\boldsymbol{\theta})$ is bounded below by a positive constant. Hence V is positive definite and radially unbounded in $(\mathbf{e}, \dot{\boldsymbol{\theta}})$. Along (7),

$$\dot{V} = -\dot{\boldsymbol{\theta}}^T \mathbf{D} \dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\theta}}^T \boldsymbol{\tau}_{\text{ext}} + \dot{\boldsymbol{\theta}}^T \big(\tfrac{1}{2}\dot{\mathbf{M}} - \mathbf{C}\big)\dot{\boldsymbol{\theta}}.$$

The last term is zero. With $\mathbf{u} = \mathbf{0}$ and $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$ the open-loop energy $E = \frac{1}{2}\dot{\boldsymbol{\theta}}^T \mathbf{M} \dot{\boldsymbol{\theta}} + P(\boldsymbol{\theta})$, with $\boldsymbol{\tau}_g = -\partial P / \partial \boldsymbol{\theta}$, has $\dot{E} = \dot{\boldsymbol{\theta}}^T (\frac{1}{2}\dot{\mathbf{M}} - \mathbf{C})\dot{\boldsymbol{\theta}}$, and E is conserved for every state. This uses only the vector $\mathbf{C}\dot{\boldsymbol{\theta}}$, so it holds for the $\mathbf{C}$ of Sec. III (Remark 1); numerically the term is below 5×10^{-9}. Thus $\dot{V} \le \dot{\boldsymbol{\theta}}^T \boldsymbol{\tau}_{\text{ext}}$, and with $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$, $\dot{V} = -\dot{\boldsymbol{\theta}}^T \mathbf{D} \dot{\boldsymbol{\theta}} \le 0$. On the set $\dot{V} = 0$ we have $\dot{\boldsymbol{\theta}} \equiv \mathbf{0}$, so $\ddot{\boldsymbol{\theta}} \equiv \mathbf{0}$ and (7) gives $\mathbf{K}\mathbf{e} = \mathbf{0}$, i.e. $\mathbf{e} = \mathbf{0}$. The largest invariant set in $\{\dot{V} = 0\}$ is the equilibrium, and LaSalle's invariance principle [5] gives global asymptotic stability. □

Two consequences matter below. First, the proof is unchanged if $\mathbf{D}$ is replaced by a symmetric matrix function $\mathbf{D}(\boldsymbol{\theta})$ whose eigenvalues are bounded below by a positive constant. Second, the inequality $\dot{V} \le \dot{\boldsymbol{\theta}}^T \boldsymbol{\tau}_{\text{ext}}$ says the closed loop is *passive* from $\boldsymbol{\tau}_{\text{ext}}$ to $\dot{\boldsymbol{\theta}}$ with storage V [5], [6]. Over any time interval the arm can return no more energy than it stored initially plus what the environment put in. The feedback connection of passive systems is again passive [5], so when the arm touches a passive environment (a spring, a damper, a mass) the energy stored in the two together cannot grow. This is the property that makes impedance control safe in contact [1].

C. Local analysis

With $\mathbf{z} = [\mathbf{e}^T \ \dot{\mathbf{e}}^T]^T$, linearizing (7) about $\mathbf{z} = \mathbf{0}$, $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$ yields

$$\dot{\mathbf{z}} = \underbrace{\begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}_d^{-1}\mathbf{K} & -\mathbf{M}_d^{-1}\mathbf{D} \end{bmatrix}}_{\mathbf{A}_{cl}} \mathbf{z} + \begin{bmatrix} \mathbf{0} \\ \mathbf{M}_d^{-1} \end{bmatrix} \boldsymbol{\tau}_{\text{ext}}, \qquad (10)$$

where $\mathbf{M}_d = \mathbf{M}(\boldsymbol{\theta}_d)$. The Coriolis term drops out because it is quadratic in the velocities, so the linearization is a mass-spring-damper with the arm's inertia and the stiffness and damping we choose. Applying the argument of Proposition 1 to this linear system with $V = \frac{1}{2}\dot{\mathbf{e}}^T \mathbf{M}_d \dot{\mathbf{e}} + \frac{1}{2}\mathbf{e}^T \mathbf{K} \mathbf{e}$ shows that $\mathbf{A}_{cl}$ is Hurwitz whenever $\mathbf{M}_d$, $\mathbf{K}$ and $\mathbf{D}$ are positive definite. For the example arm with $\boldsymbol{\theta}_d = \mathbf{0}$, $\mathbf{K} = 20\,\mathbf{I}$ and $\mathbf{D} = 5\,\mathbf{I}$, the gains of our proposal, the eigenvalues of $\mathbf{A}_{cl}$ are -24.4, -4.79 and $-0.43 \pm j1.80$. The pair $-0.43 \pm j1.80$ is lightly damped and dominates the response; Sec. V explains why.

V. Damping Design

A. Why one damping gain cannot suit both modes

Let $\mathbf{K} = k\mathbf{I}$ and $\mathbf{D} = d\mathbf{I}$, and let m_i, $\mathbf{v}_i$ be the eigenvalues and eigenvectors of $\mathbf{M}_d$. Then $\mathbf{M}_d^{-1}\mathbf{K}$ and $\mathbf{M}_d^{-1}\mathbf{D}$ share the eigenvectors $\mathbf{v}_i$, and (10) splits into two scalar systems,

$$m_i s^2 + d\, s + k = 0, \qquad \omega_{n,i} = \sqrt{k/m_i}, \qquad \zeta_i = \frac{d}{2\sqrt{k\, m_i}}. \qquad (11)$$

The ratio of the two damping ratios is $\sqrt{m_{\max}/m_{\min}}$, whatever k and d are. For the example arm $\mathbf{M}$ depends only on θ_2. With the arm straight ($\theta_2 = 0$) its eigenvalues are 0.17 and 5.83 kg·m^2, with eigenvectors $(0.41, -1)$ and $(1, 0.41)$. The heavy direction is the whole arm swinging with the elbow nearly straight. The light direction is the forearm moving against the upper arm. The ratio is 34, so the two damping ratios differ by a factor of $\sqrt{34} \approx 5.8$. With $k = 20$, $d = 5$ the forearm mode is overdamped ($\zeta = 1.35$) while the whole-arm mode rings ($\zeta = 0.23$). Raising d to fix the second overdamps the first further. With the elbow folded ($\theta_2 = \pi$) both eigenvalues equal 1 and the problem disappears, so the difficulty depends on the pose.

B. A uniform damping ratio

For any symmetric positive definite $\mathbf{K}$, choose

$$\mathbf{D} = 2\zeta\, \mathbf{M}_d^{1/2}\, \boldsymbol{\Omega}\, \mathbf{M}_d^{1/2}, \qquad \boldsymbol{\Omega} = \big(\mathbf{M}_d^{-1/2} \mathbf{K} \mathbf{M}_d^{-1/2}\big)^{1/2}. \qquad (12)$$

Substituting $\mathbf{e} = \mathbf{M}_d^{-1/2}\mathbf{q}$ into the unforced linear closed loop $\mathbf{M}_d\ddot{\mathbf{e}} + \mathbf{D}\dot{\mathbf{e}} + \mathbf{K}\mathbf{e} = \mathbf{0}$ and multiplying by $\mathbf{M}_d^{-1/2}$ gives $\ddot{\mathbf{q}} + 2\zeta\boldsymbol{\Omega}\dot{\mathbf{q}} + \boldsymbol{\Omega}^2\mathbf{q} = \mathbf{0}$. Since $\boldsymbol{\Omega}$ is symmetric positive definite, it has an orthonormal eigenbasis, and each mode obeys $s^2 + 2\zeta\omega_i s + \omega_i^2 = 0$, where ω_i^2 are the eigenvalues of $\mathbf{M}_d^{-1}\mathbf{K}$. Every mode therefore has damping ratio exactly ζ. The matrix $\mathbf{D}$ is congruent to $2\zeta\boldsymbol{\Omega}$, hence symmetric positive definite, and Proposition 1 still applies. For $\mathbf{K} = k\mathbf{I}$ the formula reduces to

$$\mathbf{D} = 2\zeta\sqrt{k}\, \mathbf{M}_d^{1/2}. \qquad (13)$$

This coincides with the double-diagonalization damping design used for Cartesian impedance control [7] when all modal damping ratios are equal. We checked numerically that (12) gives $\zeta_i = 0.7$ for all four eigenvalues with $\mathbf{K} \in \{20\mathbf{I},\ \mathrm{diag}(20, 40),\ \left[\begin{smallmatrix} 30 & 5 \\ 5 & 10 \end{smallmatrix}\right]\}$ at three poses.

For $k = 20$, $\zeta = 0.7$ and $\boldsymbol{\theta}_d = \mathbf{0}$,

$$\mathbf{D} = \begin{bmatrix} 13.28 & 4.43 \\ 4.43 & 4.43 \end{bmatrix} \ \mathrm{N \cdot m \cdot s/rad},$$

and the eigenvalues of $\mathbf{A}_{cl}$ become $-7.56 \pm j7.71$ and $-1.30 \pm j1.32$. The slowest decay rate rises from 0.43 to 1.30 s^{-1}. Because $\mathbf{M}_d$ depends on $\boldsymbol{\theta}_d$, $\mathbf{D}$ must be recomputed for each target. An alternative evaluates $\mathbf{D}(\boldsymbol{\theta}) = 2\zeta\sqrt{k}\, \mathbf{M}(\boldsymbol{\theta})^{1/2}$ online; its eigenvalues are bounded below because those of $\mathbf{M}$ are, so Proposition 1 covers it too.

VI. Simulation Results

We integrate the full nonlinear model (3) in Python with SciPy's DOP853 integrator (relative tolerance 10^{-9}, maximum step 5 ms), with $m_i = l_i = 1$ and exact gravity compensation unless stated. The model code is first checked against an independent derivation of $\mathbf{M}$ and $\boldsymbol{\tau}_g$ from kinetic and potential energy. Settling time is the last time $\|\mathbf{e}\|$ exceeds 2% of the step size, and overshoot is the largest motion past the target along the step direction.

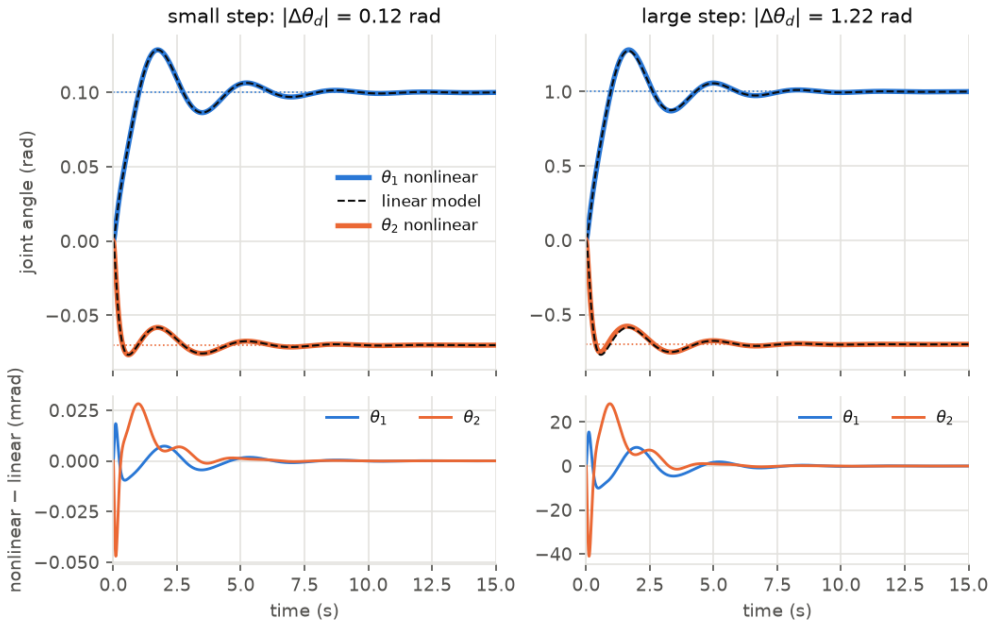


Fig. 2. Step in $\boldsymbol{\theta}_d$ with $\mathbf{K} = 20\mathbf{I}$, $\mathbf{D} = 5\mathbf{I}$. Top: joint angles from the nonlinear simulation (thick) and the linear model (dashed); dotted lines are the targets. Bottom: nonlinear minus linear, in mrad. The ringing at about 1.8 rad/s is the whole-arm mode with $\zeta = 0.23$.

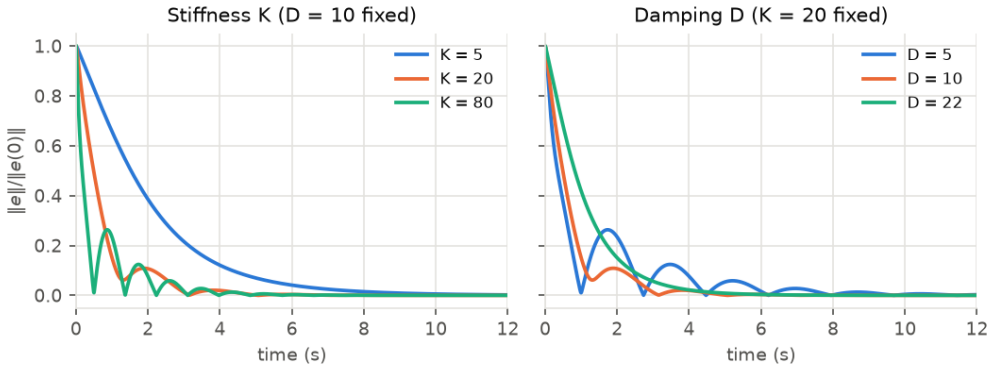


Fig. 3. Normalized error for the step of $(0.3, -0.2)$ rad with scalar gains. Left: stiffness k with $d = 10$. Right: damping d with $k = 20$.

A. Does the linear model describe the arm?

Table III compares the nonlinear simulation with the predictions of (10) and (8). To identify the eigenvalues, we fit a discrete-time linear map by least squares to a small-amplitude nonlinear trajectory and convert its eigenvalues to continuous time. They match $\mathbf{A}_{cl}$ to 1.8×10^{-3}. Steps in $\boldsymbol{\theta}_d$ (Fig. 2) agree to 0.04% of the step size for a 0.12 rad step and to 3.4% for a 1.22 rad step. The linear model stays accurate for large motions because gravity is cancelled exactly and the Coriolis torques remain small at these speeds. Under a 0.3 s push the error peaks at 0.103 rad (Fig. 4), and the linear model matches to 0.1% of the peak. Under a constant push the steady error equals $\mathbf{K}^{-1}\boldsymbol{\tau}_{\text{ext}}$ to machine precision at two poses, confirming (8).

Global stability. To test Proposition 1 away from the linear regime we started the arm from 20 random states with $|\theta_i| \le \pi$ and $|\dot{\theta}_i| \le 3$ rad/s, with $\boldsymbol{\theta}_d = (0.5, 0.5)$. For both $\mathbf{D} = 5\mathbf{I}$ and the online $\mathbf{D}(\boldsymbol{\theta})$ of Sec. V, V never increased between samples, and after 40 s every trajectory was within 10^{-7} of the equilibrium.

TABLE III

NONLINEAR SIMULATION VERSUS LINEAR PREDICTIONS ($\mathbf{K} = 20\mathbf{I}$, $\mathbf{D} = 5\mathbf{I}$ UNLESS NOTED).

Test	Result	Agreement
Eigenvalues of $\mathbf{A}_{cl}$	identified from simulation	1.8×10^{-3}
Step, 0.12 rad	settling 7.39 s (linear 7.39 s)	0.04% of step
Step, 1.22 rad	settling 6.95 s (linear 6.92 s)	3.4% of step
Push $(4, -2)$ N·m, 0.3 s	peak error 0.103 rad	0.1% of peak
Constant push, $\mathbf{K} = \text{diag}(20, 40)$	$\mathbf{e}_{ss}$ vs. $\mathbf{K}^{-1}\boldsymbol{\tau}_{\text{ext}}$	2×10^{-15}

B. Stiffness versus damping with scalar gains

Table IV and Fig. 3 show a step of $(0.3, -0.2)$ rad for $\mathbf{K} = k\mathbf{I}$, $\mathbf{D} = d\mathbf{I}$. Raising k with d fixed speeds up the response but lowers the smallest modal damping ratio, so

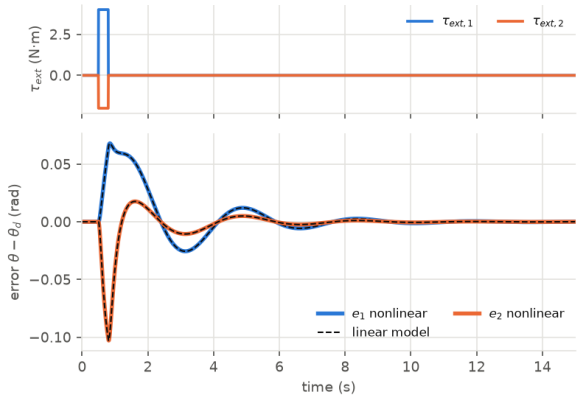


Fig. 4. Push of $(4, -2)$ N·m for 0.3 s at $\boldsymbol{\theta}_d = \mathbf{0}$, $\mathbf{K} = 20\mathbf{I}$, $\mathbf{D} = 5\mathbf{I}$. The arm yields and returns; the linear model (dashed) overlaps the nonlinear response.

overshoot grows (15% at $k = 80$, $d = 10$). Raising d with k fixed removes the overshoot, but past a point it slows the response: both modes become overdamped, and the slow pole of each approaches $-k/d$ (for $k = 5$, $d = 40$ the arm has not settled after 20 s). The smallest modal ζ in the grid tracks the overshoot closely, as (11) predicts. In every entry the two damping ratios differ by a factor of 5.7 ($\sqrt{m_{\max}/m_{\min}}$ at this target), so no scalar pair puts both modes near $\zeta = 0.7$.

TABLE IV

SCALAR GAINS, STEP OF $(0.3, -0.2)$ RAD: 2% SETTLING TIME (S) / OVERSHOOT (%) / SMALLEST MODAL ζ.

	$d = 2.5$	$d = 5$	$d = 10$	$d = 22$	$d = 40$
$k = 5$	14.8/15/0.23	7.9/5/0.47	7.4/0/0.93	17.0/0/2.05	>20/0/3.72
$k = 20$	15.4/21/0.12	7.4/15/0.23	4.0/5/0.47	4.1/0/1.02	7.7/0/1.86
$k = 80$	15.3/26/0.06	7.7/21/0.12	3.7/15/0.23	1.5/3/0.51	1.9/0/0.93

C. Uniform damping versus scalar damping

Table V and Fig. 5 compare, at $k = 20$, the proposal's $\mathbf{D} = 5\mathbf{I}$, the best scalar damping from Table IV ($\mathbf{D} = 10\mathbf{I}$), and the uniform design (13) with $\zeta = 0.7$, both fixed at $\boldsymbol{\theta}_d$ and evaluated online.

Three results stand out. First, the uniform design settles the large step in 2.6 s with 1.3% overshoot, against 7.0 s and 13% for the proposal's gains and 2.8 s and 3.9% for the best scalar gain, without a larger peak torque (20.0 versus 20.8–21.7 N·m). Second, the fixed and online versions are almost identical (2.59 versus 2.62 s), so designing $\mathbf{D}$ at the target is enough for these motions. Third, the uniform design deflects *more* under the push: 0.149 rad, against 0.103 and 0.063 rad. The push mostly excites the light forearm direction, where the scalar designs are heavily overdamped ($\zeta = 1.35$ and 2.7) and therefore resist fast motion strongly; the uniform design damps that direction only to $\zeta = 0.7$. It still settles fastest after the push (2.7 s versus 8.8 and 5.5 s). The damping matrix is part of the impedance the environment feels, so choosing it for a good step response also fixes how hard the arm resists a sudden push.

D. Error in the gravity model

The law (6) cancels gravity with a model. If the controller believes $m_2 = 0.9$ kg while the true value is 1 kg, the arm settles at $\boldsymbol{\theta}_d = (1.0, -0.7)$ with error $(-0.051, -0.012)$ rad for $k = 20$ and $(-0.014, -0.003)$ rad for $k = 80$. The first-order estimate $\mathbf{K}^{-1}(\boldsymbol{\tau}_g - \hat{\boldsymbol{\tau}}_g)$, from (8) with the gravity error acting as an external torque, gives $(-0.056, -0.015)$ and $(-0.014, -0.004)$ rad. The error falls as $1/k$, so a soft arm needs an accurate gravity model.

VII. DISCUSSION AND LIMITATIONS

Model knowledge. The law cancels gravity exactly only if the model is exact. Sec. VI shows that a 10% mass error leaves a steady error proportional to $1/k$. Integral action would remove it, but it would also change the impedance the environment feels, since a constant push would then be rejected instead of yielded to. The uniform damping design additionally needs $\mathbf{M}(\boldsymbol{\theta}_d)$.

Measurement. We assumed both angles and velocities are measured without noise. Sec. III-D shows that the angles alone, or even one sensor, make the linearization observable, so an observer could supply the velocities; we did not design one.

Physical effects left out. The model has no friction, no actuator limits and no delays. The peak motor torques in our tests are about 22 N·m, and friction would add damping that the design would have to account for.

Joint space. The environment pushes on the hand, but our stiffness is defined at the joints. For small motions about a nonsingular pose with Jacobian $\mathbf{J}$, a joint stiffness $\mathbf{K}$ appears at the hand as $\mathbf{J}^{-T}\mathbf{K}\mathbf{J}^{-1}$, which varies with the pose. A Cartesian impedance law would choose the hand stiffness directly.

Constant target. We regulate to a constant $\boldsymbol{\theta}_d$. Following a moving target would need feedforward of $\mathbf{M}\ddot{\boldsymbol{\theta}}_d + \mathbf{C}\dot{\boldsymbol{\theta}}_d$.

VIII. CONCLUSION

We modeled a two-link arm in state-space form and showed that its hanging equilibrium is only marginally stable, since without friction it swings forever. Its linearization is controllable and observable at every equilibrium, and it stays so with one motor or one sensor as long as gravity couples the joints. Since no open-loop mode decays on its own, stabilizability coincides with controllability here. The impedance law turns the arm into a mass-spring-damper with a static stiffness we choose exactly. An energy argument proves global asymptotic stability and passivity, and it holds for our Coriolis matrix even though that matrix is not the skew-symmetric choice. Because the arm's inertia is 34 times larger in one direction than the other, a scalar damping gain always leaves one mode poorly damped. A damping matrix built from $\mathbf{M}^{1/2}$ gives every mode the same damping ratio and cuts the settling time of a large step from 7.0 s to 2.6 s, at the price of less resistance to sudden pushes. Every prediction was confirmed against simulations of the full nonlinear model.

TABLE V

SCALAR VERSUS UNIFORM DAMPING ($\mathbf{K} = 20\mathbf{I}$), NONLINEAR SIMULATION. STEPS START FROM REST AT $\boldsymbol{\theta} = \mathbf{0}$; THE PUSH IS APPLIED AT $\boldsymbol{\theta}_d = \mathbf{0}$.

	Step to $(1.0, -0.7)$ rad		Step to $(0.3, -0.2)$ rad	Push $(4, -2)$ N·m, 0.3 s	
Damping	settling (s) / overshoot	peak $\lvert u_i \rvert$ (N·m)	settling (s) / overshoot	peak $\lvert e_i \rvert$ (rad)	settling (s)
$\mathbf{D} = 5\mathbf{I}$ (proposal)	6.95 / 13.2%	21.7	7.38 / 14.6%	0.103	8.77
$\mathbf{D} = 10\mathbf{I}$	2.81 / 3.9%	20.8	3.96 / 4.8%	0.063	5.54
Uniform, $\zeta = 0.7$, fixed at $\boldsymbol{\theta}_d$	2.59 / 1.3%	20.0	2.79 / 1.4%	0.149	2.71
Uniform, $\zeta = 0.7$, online $\mathbf{D}(\boldsymbol{\theta})$	2.62 / 1.3%	20.0	2.79 / 1.4%	0.149	2.71

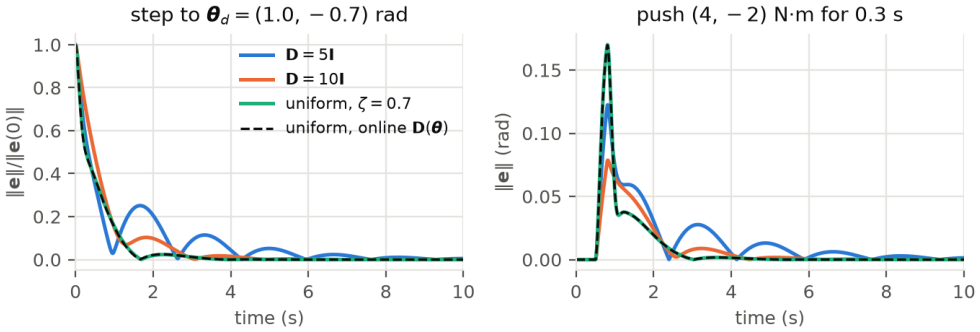


Fig. 5. Scalar versus uniform damping with $\mathbf{K} = 20\mathbf{I}$. Left: step to $(1.0, -0.7)$ rad. Right: push of $(4, -2)$ N·m for 0.3 s. The uniform design removes the slow ringing of the whole-arm mode but lets the arm deflect further under the push.

REFERENCES

[1] N. Hogan, "Impedance control: An approach to manipulation: Part i—theory," *Journal of Dynamic Systems, Measurement, and Control*, vol. 107, no. 1, pp. 1–7, 03 1985. [Online]. Available: https://doi.org/10.1115/1.3140702

[2] M. Takegaki and S. Arimoto, "A new feedback method for dynamic control of manipulators," *Journal of Dynamic Systems, Measurement, and Control*, vol. 103, no. 2, pp. 119–125, 1981.

[3] R. Tedrake, *Underactuated Robotics*, 2023. [Online]. Available: https://underactuated.csail.mit.edu

[4] C.-T. Chen, *Linear System Theory and Design*, 3rd ed. New York: Oxford University Press, 1999.

[5] H. K. Khalil, *Nonlinear Systems*, 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2002.

[6] M. W. Spong, S. Hutchinson, and M. Vidyasagar, *Robot Modeling and Control*. John Wiley & Sons, 2006.

[7] A. Albu-Schäffer, C. Ott, U. Frese, and G. Hirzinger, "Cartesian impedance control of redundant robots: Recent results with the DLR-light-weight-arms," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, Taipei, Taiwan, 2003, pp. 3704–3709.